

***Inteligência Artificial
e Computacional***

**ELT578
ANÁLISE DE IMAGENS E VISÃO COMPUTACIONAL**

Aula 9:

Segmentação de Imagem via CNN e Descritores de Textura

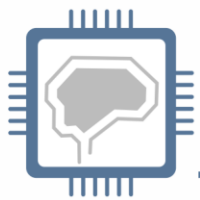
Conteudista: M.Sc. Talita E. Z. Santana



CONTEÚDO

SEMANA 03: COR, SEGMENTAÇÃO E ANÁLISE DE IMAGENS.

- Cor,
- Percepção da Cor
- Brilho, intensidade e luminância
- Espaço de Cores
- Segmentação
 - Limiarização
 - Detecção de Borda
 - CNN
- Introdução a Análise de Imagens
 - Descritores de Objeto

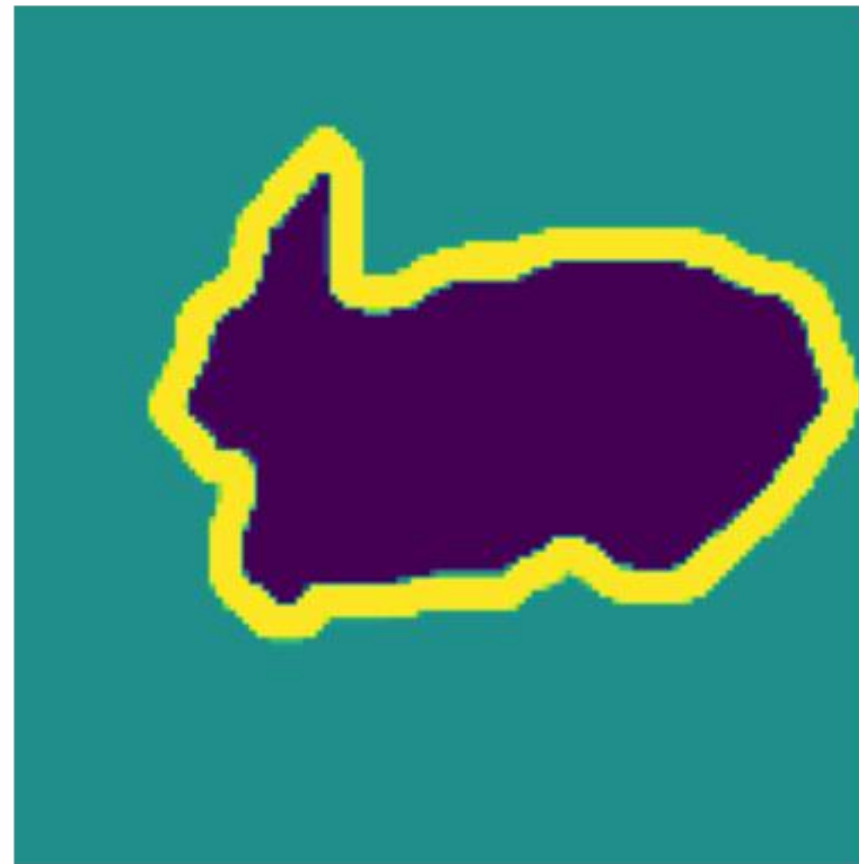


Segmentação de Imagens via CNN

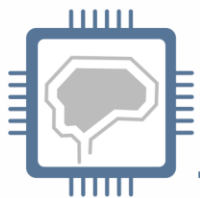
Input Image



True Mask



- **Classe 1:** Pixel pertencente ao animal de estimação.
- **Classe 2:** Pixel na fronteira do animal de estimação.
- **Classe 3:** Nenhuma das opções acima/ Pixel circundante.



Segmentação de Imagens via CNN



MobileNet



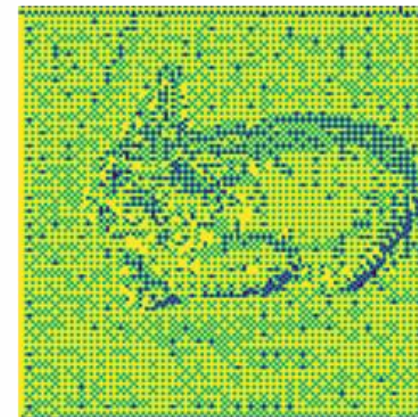
Input Image



True Mask



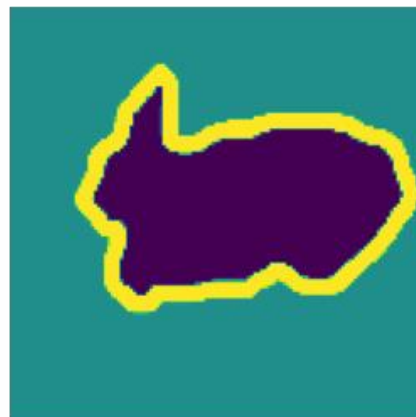
Predicted Mask



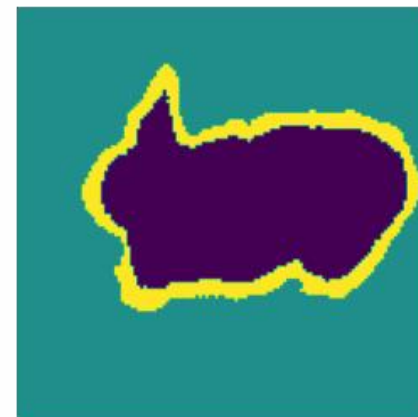
Input Image



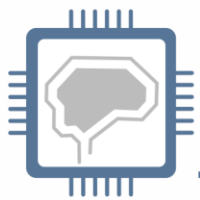
True Mask



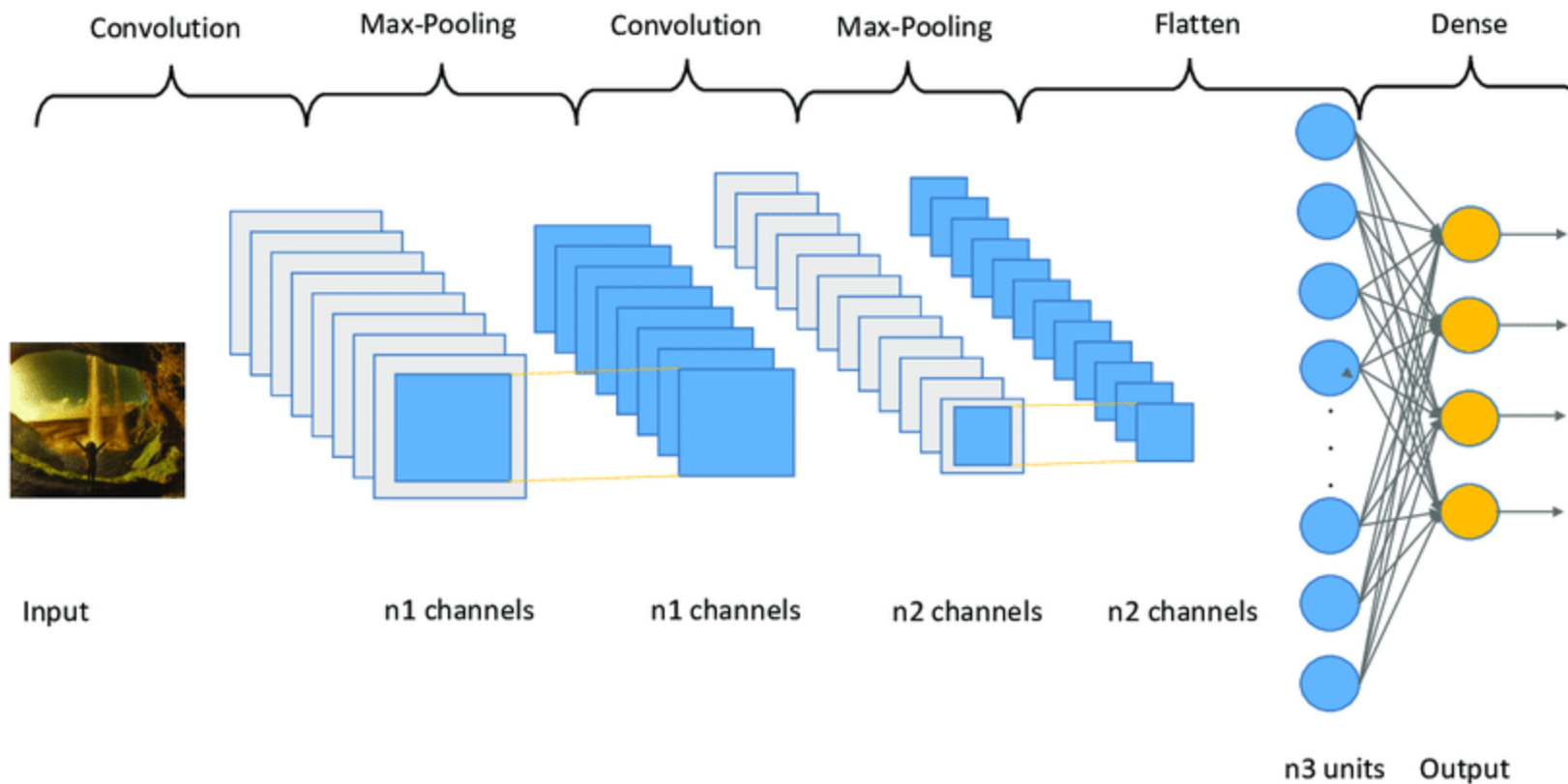
Predicted Mask

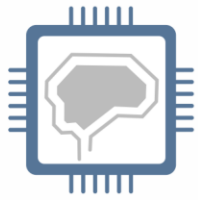


- **Classe 1:** Pixel pertencente ao animal de estimação.
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Segmentação de Imagens via CNN

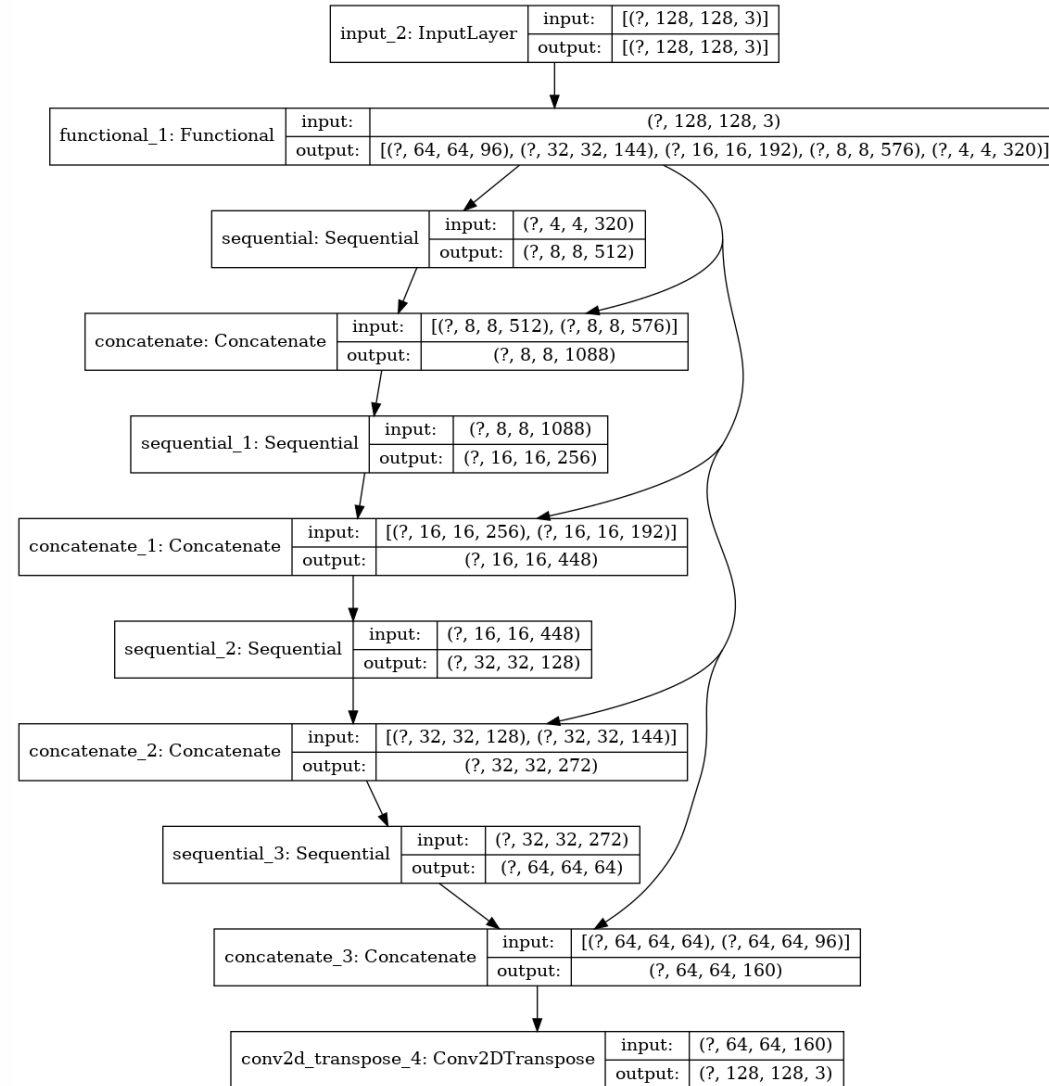


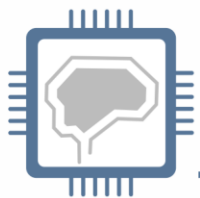


Segmentação de Imagens via CNN



MobileNet

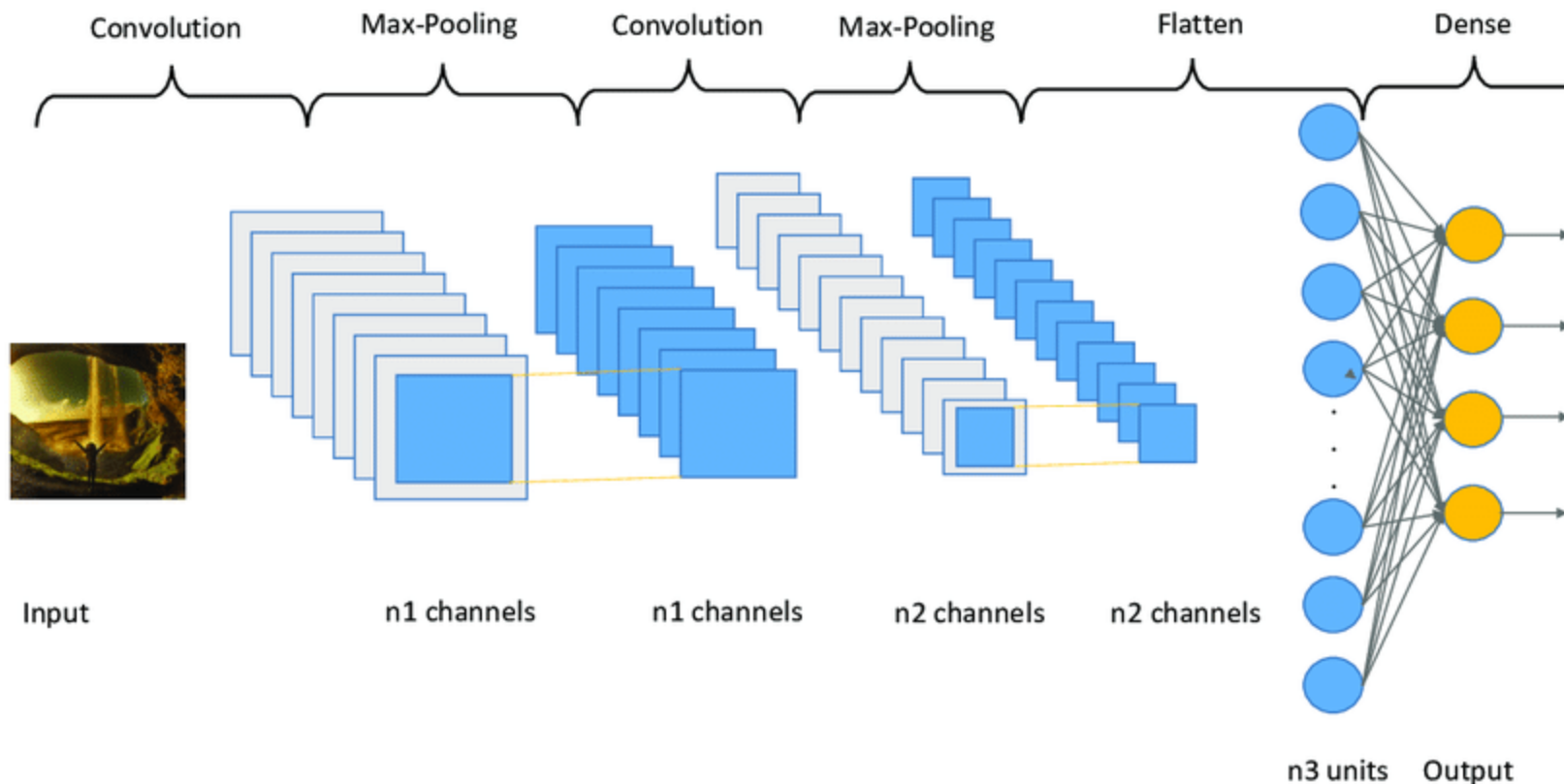


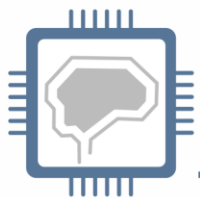


Segmentação de Imagens via CNN



MobileNet

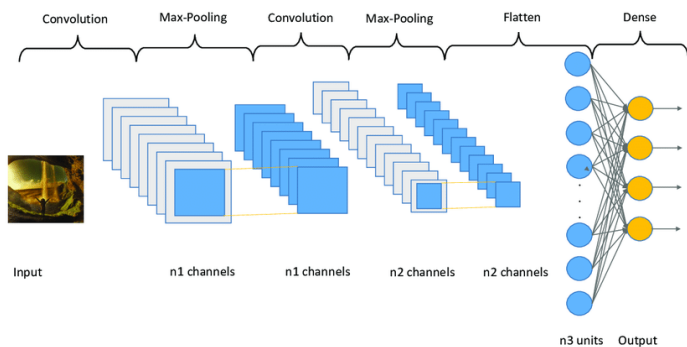




Segmentação de Imagens via CNN



MobileNet



Input Volume (+pad 1) (7x7x3)

$x[:, :, 0]$

0	0	0	0	0	0	0
0	0	0	1	0	2	0
0	1	0	2	0	1	0
0	1	0	2	2	0	0
0	2	0	0	2	0	0
0	2	1	2	2	0	0
0	0	0	0	0	0	0

$x[:, :, 1]$

0	0	0	0	0	0	0
0	2	1	2	1	1	0
0	2	1	2	0	1	0
0	0	2	1	0	1	0
0	1	2	2	2	2	0
0	0	1	2	0	1	0
0	0	0	0	0	0	0

$x[:, :, 2]$

0	0	0	0	0	0	0
0	2	1	1	2	0	0
0	1	0	0	1	0	0
0	0	1	0	0	0	0
0	1	0	2	1	0	0
0	2	2	1	1	1	0
0	0	0	0	0	0	0

Filter W0 (3x3x3)

$w0[:, :, 0]$

-1	0	1
0	0	1
1	-1	1

$w0[:, :, 1]$

-1	0	1
1	-1	1
0	1	0

$w0[:, :, 2]$

-1	1	1
1	1	0
0	-1	0

Bias $b0$ (1x1x1)

$b0[:, :, 0]$

1

Filter W1 (3x3x3)

$w1[:, :, 0]$

0	1	-1
0	-1	0
0	-1	1

$w1[:, :, 1]$

-1	0	0
1	-1	0
1	-1	0

$w1[:, :, 2]$

-1	1	-1
0	-1	-1
1	0	0

Bias $b1$ (1x1x1)

$b1[:, :, 0]$

0

toggle movement

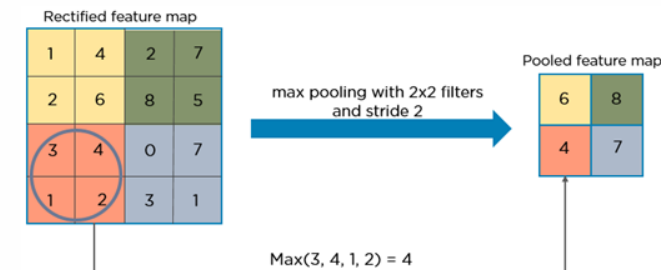
Output Volume (3x3x2)

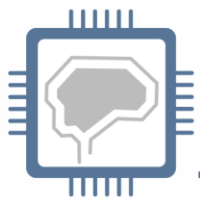
$o[:, :, 0]$

2	3	3
3	7	3
8	10	-3

$o[:, :, 1]$

-8	-8	-3
-3	1	0
-3	-8	-5





Convolução

$u, v \in [-1, 1]$

$h(-1, -1) = a$

$h(-1, 0) = b$

$h(-1, 1) = c$

.

.

.

$h(1, 1) = i$

a (-1,-1)	b (-1,0)	c (-1,1)
d (0,-1)	e (0,0)	f (0,1)
g (1,-1)	h (1,0)	i (1,1)

$h(u, v)$

A (i-1,j-1)	B (i-1,j)	C (i-1,j+1)
D (i,j-1)	E (i,j)	F (i,j+1)
G (i+1,j-1)	H (i+1,j)	I (i+1,j+1)

$f(i, j)$

$f(i-1, j-1) = A$

$f(i-1, j) = B$

$f(i-1, j+1) = C$

.

.

.

$f(i+1, j+1) = I$

$$g(i, j) = \sum_{u=-1}^{+1} \sum_{v=-1}^{+1} [h(u, v) \cdot f(i-u, j-v)]$$

$u = -1$

* $v = -1 \rightarrow h(-1, -1) \cdot f(i+1, j+1) = a \cdot I$

* $v = 0 \rightarrow h(-1, 0) \cdot f(i+1, j) = b \cdot H$

* $v = 1 \rightarrow h(-1, 1) \cdot f(i+1, j-1) = c \cdot G$

$u = 0$

* $v = -1 \rightarrow h(0, -1) \cdot f(i, j+1) = d \cdot F$

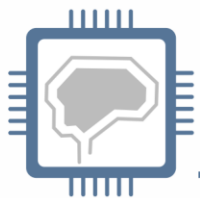
* $v = 0 \rightarrow h(0, 0) \cdot f(i, j) = e \cdot E$

* $v = 1 \rightarrow h(0, 1) \cdot f(i, j-1) = f \cdot D$

$u = 1$

...

$$g(i, j) = a \cdot I + b \cdot H + c \cdot G + d \cdot F + \dots + i \cdot A$$



Segmentação de Imagens via CNN



MobileNet



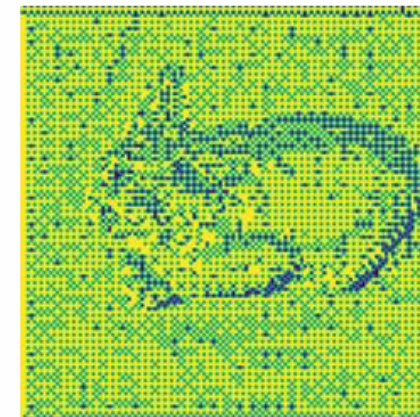
Input Image



True Mask



Predicted Mask



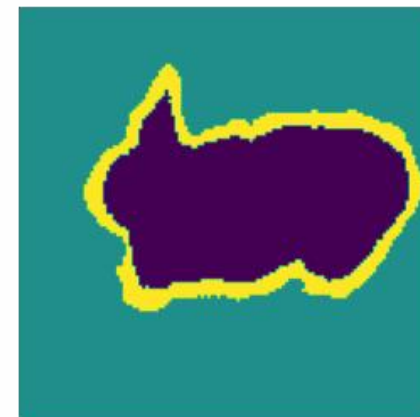
Input Image



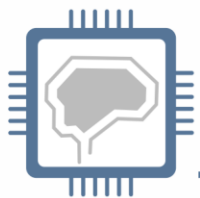
True Mask



Predicted Mask



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Descritores de Objeto - Forma



Área

Perímetro

Comprimento

Compacidade

Excetricidade

Retangularidade

Centroide

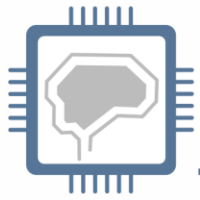
Orientação

Fator de
Forma

Contorno
Convexo

Solidez





Contraste

Dissimilaridade

Homogeneidade

Energia

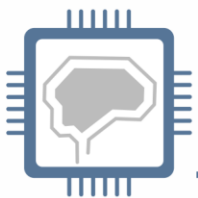
Correlação

Entropia

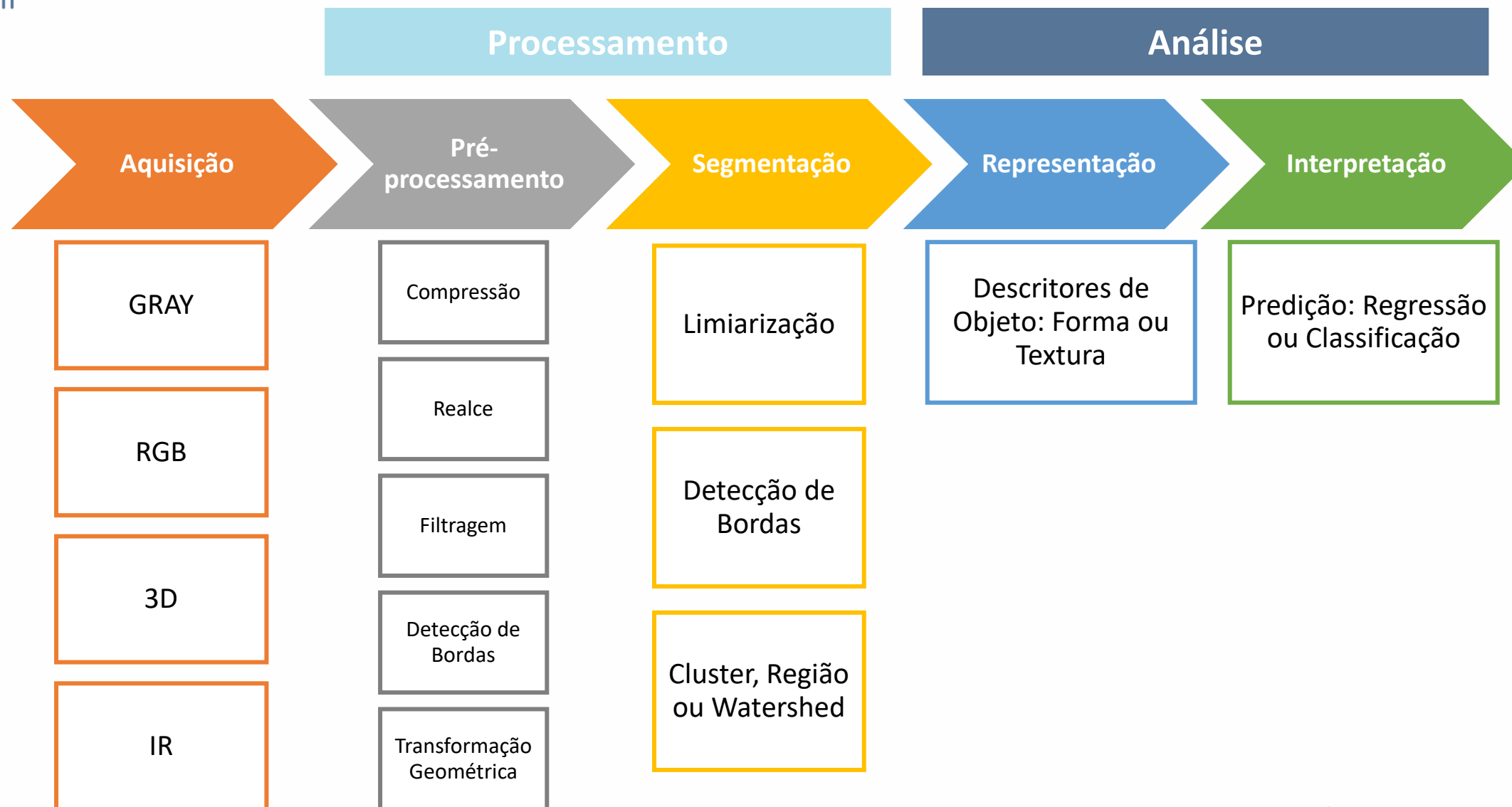
Segundo
Momento da
Média

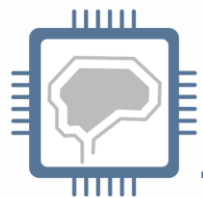
Local Binary
Patterns
(LBP):



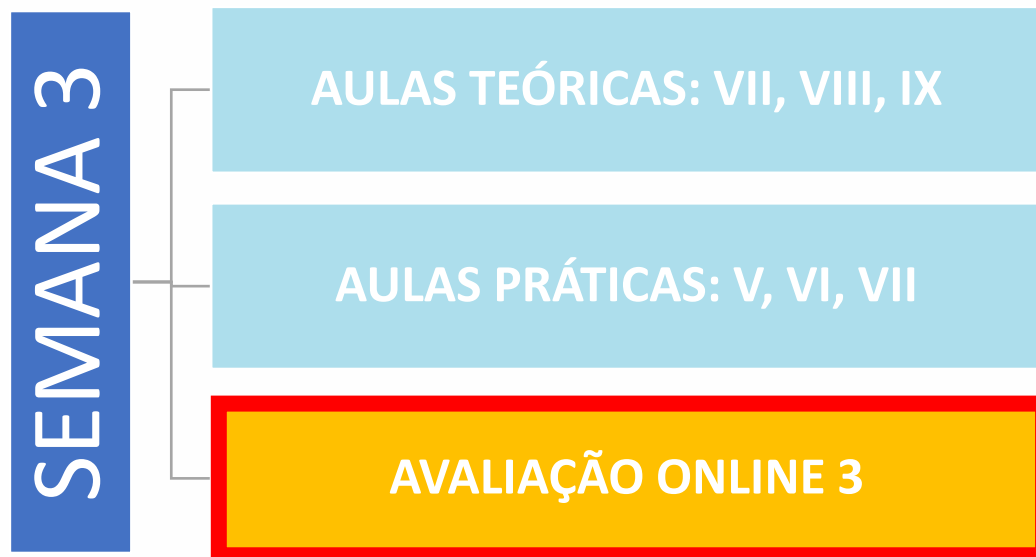


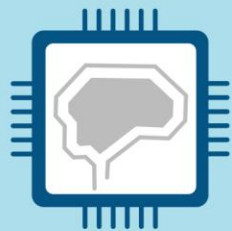
Sistema para processamento





Avaliações





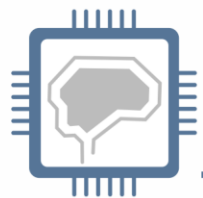
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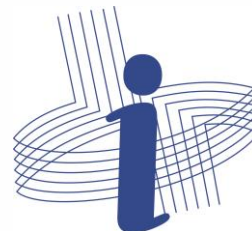
Realização

UFV

Universidade Federal de Viçosa

ENGENHARIA
ELÉTRICA

Universidade Federal de Viçosa



NIAS

Núcleo Interdisciplinar de Análise de Sinais

